Practice Final Exam — Biol 336 2019/2020 Term 2

- Due No due date
- Points 62
- Questions 20
- Available after Nov 29 at 2p.m.
- Time Limit 220 Minutes
- Allowed Attempts Unlimited

Instructions

This is the 2019/2020 spring term exam. Not all of the original questions apply this term, but some were revised to give you a sense of the type of exam questions you might see. Also, some are open-answered questions and are not automatically graded. For these, discuss with classmates and/or post your answers to Piazza to share ideas and get feedback.

Take the Quiz Again

Attempt History

	Attempt	Time	Score
KEPT	Attempt 2	285 minutes	20 out of 62 *
LATEST	Attempt 2	285 minutes	20 out of 62 *
	Attempt 1	220 minutes	3 out of 62 *

* Some questions not yet graded

* Some questions not yet graded

Submitted Dec 10 at 11:21a.m.

Comprehensive: This section is for questions whose subjects span the whol course or are from before Reading Week.

Correct answer

Question 1

1 / 1 pts

If you were to make a list of all the species described by biologists, and picked one from the list randomly, chances are it would be a animal other than vertebrate and dioecious sexual .

Answer 1:

fungus

plant

vertebrate

animal other than vertebrate

microbe

Answer 2:

hermaphrodite sexual

dioecious sexual

asexual

Correct answer

Question 2

6 / 6 pts

Match the word or phrase on the left with the phrase on the right that best matches as synonyms, cause-and-effect, logical implication, or part of the same evolutionary process. No answer is reused.

Fisher's runaway

gametic disequilibrium

convergence

homoplasy

monophyly

clade

key innovation

adaptive radiation

allele fixation

complete coalescence

sperm and egg

anisogamy

Other Incorrect Match Options:

- honest signal
- mating barrier
- predator avoidance
- heterozygote disadvantage
- behavioural plasticity

Question 3
4 / 5.5 pts

In each of these sentences, choose either "increase(s)" or "decrease(s)" to fill in the blank and generate a statement that is, in most instances, true.

A. When selection varies across space, gene flow decreases local adaptation.

B. The presence of transposable elements decreases the fitness of multicellular animals.

C. Small population size increases the probability of discord between gene tree and species tree.

D. Small population size will likely decrease the effectiveness of selection.

E. High additive genetic variation increases the chances of adaptation.

F. The rate of change in the frequency of a disadvantageous allele decreases as the favoured allele approaches fixation.

G. Modular development in animals increases pleiotropy in genes controlling functional units.

H. More evolutionary changes implied by a phylogenetic hypothesis decreases our confidence in it compared to another hypothesis with fewer, according to the criterion of parsimony.

I. High predation in juvenile life stages selects for decreases in age of sexual maturity.

J. Low extrinsic mortality in adult individuals selects to decrease the rate of senescence.

K. If we stand on the tree of life at the MRCA of spiders and humans, and move to the MRCA of echinoderms and humans, the age in millions of years increases .

Answer 1:
increases
decreases
Answer 2:
increases
decreases
Answer 3:
increases
decreases
Answer 4:
increase
decrease
Answer 5:
increases
decreases
Answer 6:
increases
decreases
Answer 7:

increases

decreases

Answer 8:

increases

decreases

Answer 9:

increases

decreases

Answer 10:

increase

decrease

Answer 11:

increases

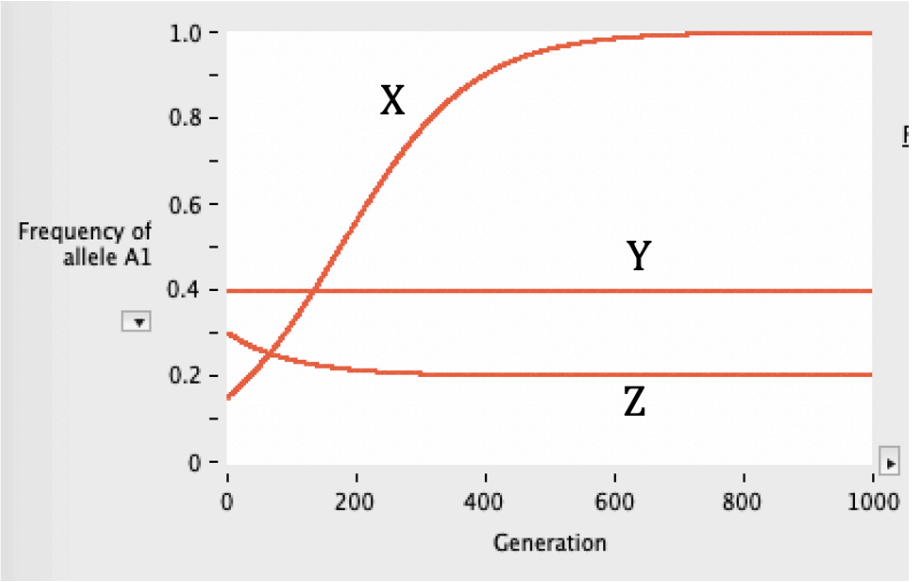
decreases

Correct answer

Question 4

5 / 5 pts

The graph below shows three simulations of the change in the frequency of an allele, A1 (labeled X, Y and Z), in an infinite population. Each was generated under a different set of parameters shown below. Note that each line had a different starting frequency of A1, simply to avoid cluttering the figure.



Match the parameters to the simulations, filling in each blank with the letter for the right simulation (X, Y or Z). Only relevant parameter values are described (other parameter don't vary).

Each graph (X, Y, Z) is the correct answer for just one scenario (two scenarios have 'no line matches this scenario' as the correct answer). Note that the notations for relative fitnesses below are

shortened (w_{11} is the relative fitness of A1A1 homozygotes, for example).

A. No selection; no mutation. [Select]

B. No selection; mutation to A1 is equal to mutation A2. [Select]

C. No selection; mutation to A1 is 4 times smaller than mutation to A2. [Select]

D. w_{11}=1.02, w_{12}=1.01, w_{22}=1.0; no mutation. [Select]

E. w_{11}=1.0, w_{12}=1.01, w_{22}=1.02; no mutation. No line matches this scenario

Answer 1:

X

Y

Z

No line matches this scenario

Answer 2:

X

Y

Z

No line matches this scenario

Answer 3:

X

Y

Z

No line matches this scenario

Answer 4:

X

Y

Z

No line matches this scenario

Answer 5:

X

Y

Z

No line matches this scenario

Correct answer

Question 5

1 / 1 pts

Place in the correct order from **oldest to youngest**, left to right. We'll use the abbreviations provided in **bold**.

- **Hum-Den**: MRCA of humans and Denisovans
- **Hum-Frog**: MRCA of humans and frogs
- **540** million years ago
- **66** million years ago
- **Present** day

- ○ Hum-Den, Hum-Frog, 540, 66, present
- ○ Hum-Den, 540, Hum-Frog, 66, present
- ◉ 540, Hum-Frog, 66, Hum-Den, present
- ○ Hum-Frog, 540, 66, Hum-Den, present
- ○ 540, Hum-Den, Hum-Frog, 66, present

Unanswered

Question 6
0 / 6 pts

In the Populations section of the course, we discussed **natural selection** changing the frequencies of alleles. In other parts of the course, we discussed **species selection**. Many statements made by biologists imply the action of one of these two processes. For each of the following statements, please indicate whether the statement is claiming or implying the action of natural selection, of species selection, or neither.

A. Most animals are sexual because it allows them to avoid Muller's Ratchet. [Select]

B. Bats evolved echolocation because of its advantage for flying at night. [Select]

C. Allele frequencies can change rapidly during a bottleneck. [Select]

D. Plant lineages with bilaterally symmetrical flowers diversify more quickly. [Select]

E. Cis-regulatory regions play a major role in controlling animal development. [Select]

F. Population comparisons show unusually high F_{ST} for the *XNF* locus compared to neutral loci, suggesting local adaptation. [Select]

Answer 1:
(You left this blank)
natural selection
species selection
neither
Answer 2:

(You left this blank)
natural selection
species selection
neither

Answer 3:

(You left this blank)
natural selection
species selection
neither

Answer 4:

(You left this blank)
natural selection
species selection
neither

Answer 5:

(You left this blank)
natural selection
species selection
neither

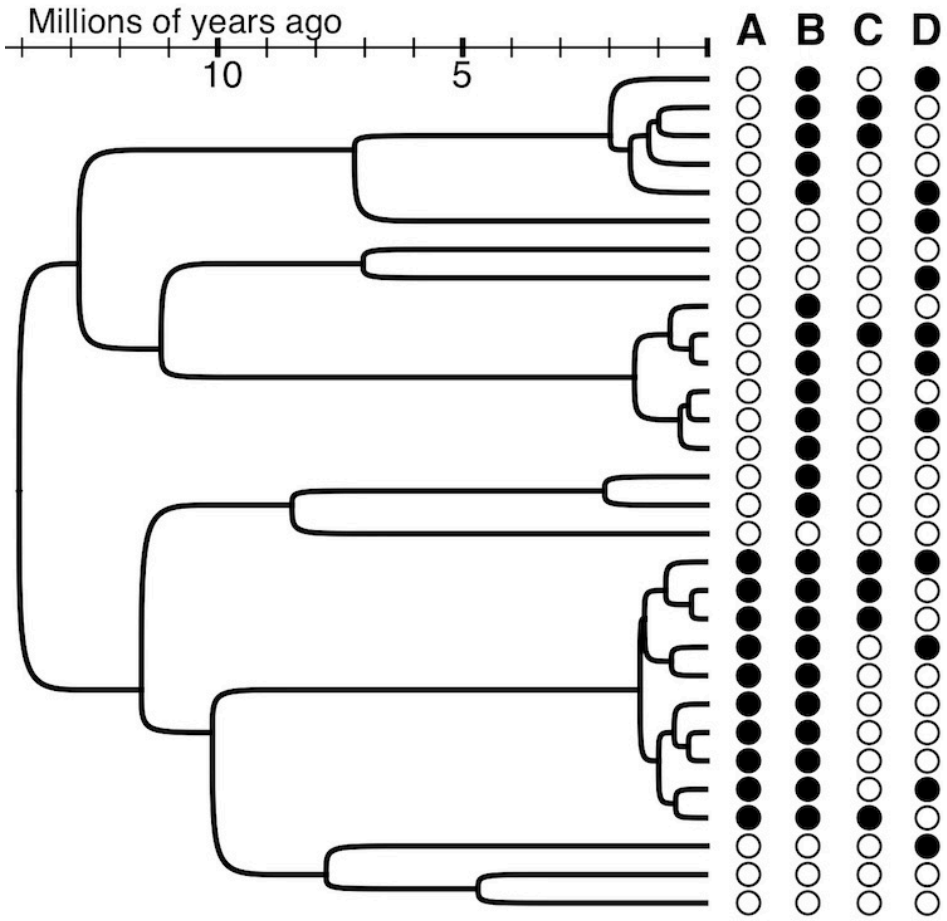
Answer 6:

(You left this blank)
natural selection
species selection
neither

Wrong answer

Question 7
0 / 2 pts

Here is a phylogeny of 30 species, shown with four different characters (A, B, C, D) that were stu
in these species. Each character has two states, distinguished by white and black dots.



The data shown provide evidence that one of the characters plays a role in controlling the rate of diversification. Which character? D

A method one can use to assess this evidence is sister species comparisons .

Answer 1:

A

B

C

D

Answer 2:

sister species comparisons

sister character comparisons

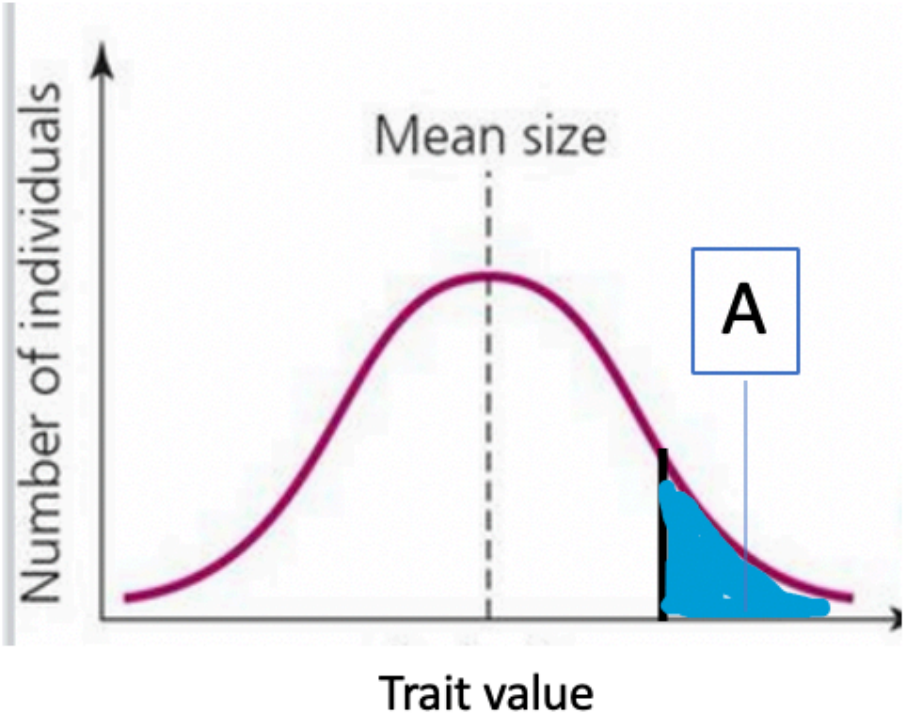
sister clade comparisons

sister allele comparisons

Questions after this are not part of the comprehensive section; they focus on material after reading week

Question 8

3 / 4 pts



The graph above shows the distribution and mean size for individuals in a population before selection. Answer the following questions about the graph.

A. What is likely indicated by the letter A? The selection differential

B. In addition to the information shown on this graph, what information would you need to calcula the heritability of this trait? The difference in mean trait value of original population (before selection) and offspring

C. If the heritability of this trait is zero, what is the expected mean trait value of the offspring? The value of the original population (before selection).

D. If the heritability of this trait is one, what is the expected mean trait value of the offspring? The trait value of the parents (after selection).

Answer 1:

The selection differential

The response to selection

The mean trait value after selection

The mean offspring trait value

Answer 2:

Breeding value of the population

The difference in mean trait value of parents (after selection) and offspring

The difference in mean trait value of original population (before selection) and offspring

The variance due to dominance interactions of alleles

Answer 3:

The trait value of the original population (before selection).

The trait value of the parents (after selection).

Neither of the above.

Answer 4:

The trait value of the original population (before selection).

The trait value of the parents (after selection).

Neither of the above.

Unanswered

Question 9

Not yet graded / 4.5 pts

As you've been sitting at home over the last few weeks, you've noticed several jumping spiders up on the ceiling. It's obvious that these spiders didn't evolve in houses, so you've been wondering if this climbing behaviour is an adaptation.

You have come up with a hypothesis: **Spiders that climb have higher survival because they avoid predators that feed on the ground.**

List and briefly describe 3 types of evidence that would support your hypothesis. Please number and separate these 3 points.

You should be able to answer in less than 75 words. .

Your Answer:

Unanswered

Question 10

0 / 4 pts

[This does not apply this term but might be of interest]

According to the mutation accumulation theory of senescence, a mutation that [Select] late in life eventually achieves fixation through [Select] . Whether or not such mutations achieve fixation depends on [Select] mortality of [Select] .

Which of these choices has a different answer in the antagonistic pleiotropy theory than in the mutation accumulation theory? [Select]

Answer 1:

(You left this blank)

occurs

acts

Answer 2:

(You left this blank)

drift

selection

Answer 3:

(You left this blank)

intrinsic

extrinsic

Answer 4:

(You left this blank)

juveniles

adults

Answer 5:

(You left this blank)

occurs vs. acts

drift vs. selection

intrinsic vs. extrinsic

juveniles vs. adults

Unanswered

Question 11

Not yet graded / 3 pts

REVISED FOR THIS TERM:

Give a biological example of cooperation between species and provide one plausible mechanism that stabilizes the cooperation against the invasion of cheating strategies.

You should be able to answer this is in 75 words or less.

[ORIGINAL QUESTION THAT DOES NOT APPLY THIS TERM]

Give an example of frequency-dependent selection, and explain its mechanism (including how th selection depends on frequency) and what its expected outcome is.

You should be able to answer this is in 75 words or less.

Your Answer:

Unanswered

Question 12

Not yet graded / 2 pts

In the space below describe the one important similarity and one important difference between a hybrid species and hybrid speciation. Be specific.

You should be able to answer this is in 60 words or less.

Your Answer:

Unanswered

Question 13
Not yet graded / 2 pts

REVISED FOR THIS TERM:

Pathogens can evolve to be more virulent, despite the fact that this virulence kills their host, because there are trade-offs or because of pleiotropy. If an infectious disease evolves to increase the chance of heart attacks late in life (much later than the infection), discuss whether this is more likely to have evolved as a result of trade-offs or pleiotropy and justify your answer.

You should be able to answer this is in 60 words or less. (Note that its point value.)

[ORIGINAL QUESTION THAT DOES NOT APPLY THIS TERM]

In the space below describe the one important similarity and one important difference between reinforcement (of reproductive isolation) and BDM (Bateson-Dobzhansky-Muller) post-zygotic incompatibilities.

You should be able to answer this is in 60 words or less. (Note that its point value.)

Your Answer:

Unanswered

Question 14
Not yet graded / 2 pts

In the space below describe the one important similarity and one important difference between isolation by distance and allopatric speciation.

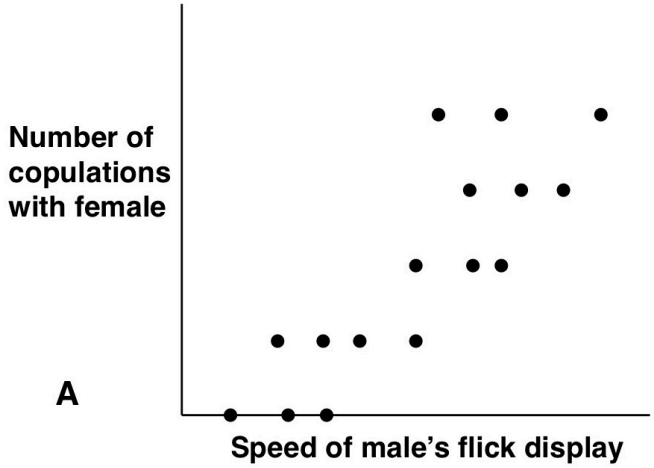
You should be able to answer this is in 60 words or less.

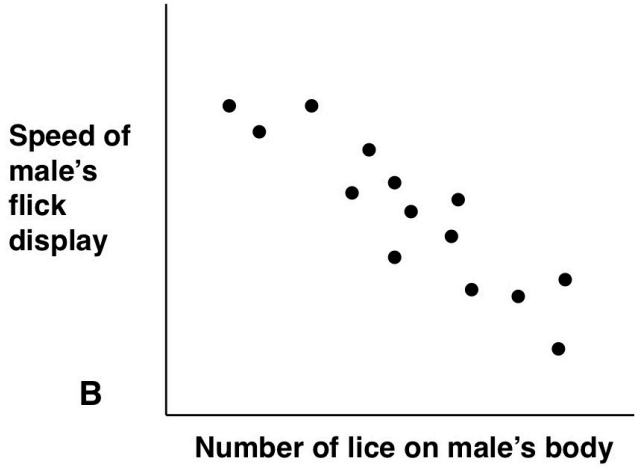
Your Answer:

Unanswered

Question 15
Not yet graded / 2 pts

Males of the blue billiballa bird do a mating dance in which the wings are flicked outward quickly. Females are likely to accept the male for mating, but vary in how many times they allow him to copulate — up to 4 times, and the more times copulating gives that male a higher chance to fertilize her eggs.





In Graph A is shown data where each spot represents an encounter between a male and a female, showing how many copulations they had depending on the speed of his flick. In Graph B is shown how the speed of his flick varies with the number of parasitic lice on his body, which are too small to be visible to the female.

What model (kind, type) of intersexual selection do these graphs suggest, and why? *(You should be able to answer this in 75 words or less — note its point value.)*

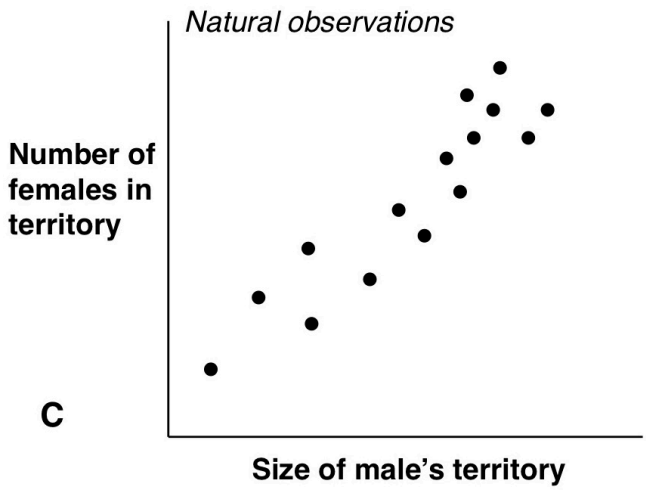
Your Answer:

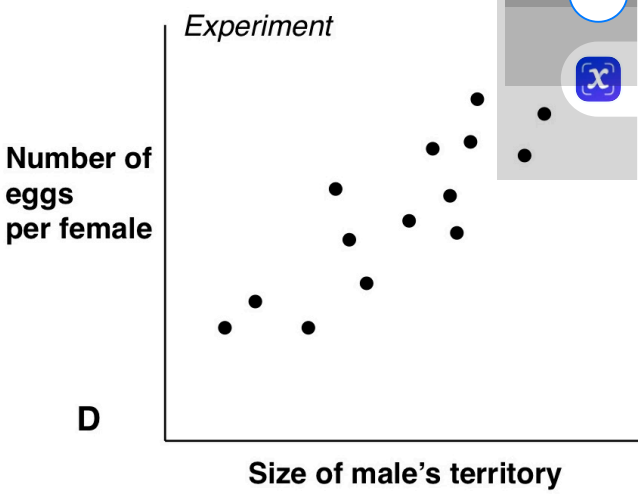
Unanswered

Question 16

Not yet graded / 2 pts

Males of the red jubjub bird establish territories, and females come to establish nests and mate with the male owning the territory. More competitive males get bigger territories. Females entering the territory establish the nest immediately, then begin to forage to prepare their bodies to make eggs. Females on a territory feed and raise offspring alone, without the male's help. Bigger territories o greater food resources for use by both the male and females.





In Graph C is shown data from observations in a natural habitat after females have established nests, but before foraging and mating has happened. At that point, the biologist caught females on nests in

various territories and randomly reassigned them to a male territory, likely not the one they had chosen. Graph D shows the number of eggs in those females that were reshuffled among territories.

What model (kind, type) of intersexual selection do these graphs suggest, and why? *(You should be able to answer this in 75 words or less — note its point value.)*

Your Answer:

Unanswered

Question 17
Not yet graded / 1 pts

REVISED FOR THIS TERM:

Discuss how evolution on another planet might differ If cooperation between species cannot be stabilized (e.g., perhaps organisms in the environment mix so fast that sanctions, reciprocity, etc. don't work). Be specific about one major difference between what you predict would have evolved/not evolved on this alternative planet. *(You should be able to answer this in 75 words or less. Note that it is only one point.)*

[ORIGINAL QUESTION THAT DOES NOT APPLY THIS TERM]

Let's suppose you find a species not with two sexes (male and female), but with three sexes that we will call flam, mef, and lemf. Each produces a distinctive type of gamete. All gametes are the same size, but to produce a zygote successfully, the two gametes that join have to be from two different sexes — two mefs can't make a baby together, for instance, but a mef can make a baby with either a flam or a lemf, and a flam can make a baby with either a mef or a lemf, and so on. You go to the population and count the frequency of mefs, flams and lemfs. What do you expect the frequencies to be, and why? *(You should be able to answer this in 75 words or less. Note that it is only one point.)*

Your Answer:

Unanswered

Question 18
0 / 3 pts

Select a species concept that is consistent with each statement. There may be more than one appropriate concept - choose any correct match. You can re-use the same answer for different statements.

A species concept that can be applied to fossils: [Select]

A species concept that would allow good species to sometimes hybridize: [Select]

A species concept that would have difficulty dealing with multiple origins of the same species: [Select]

A species concept that can accommodate two species that have the same morphology (cryptic species): [Select]

The likely basis for most species described before 1940: [Select]

A species concept that you might test using a reciprocal transplant experiment: [Select]

Answer 1:
(You left this blank)
Biological Species Concepy
Ecological Species Concept
Phylogenetic Species Concept
Genotypic Cluster Species Concept
Taxonomic Species

Answer 2:
(You left this blank)
Biological Species Concept
Ecological Species Concept
Phylogenetic Species Concept
Genotypic Cluster Species Concept
Taxonomic Species

Answer 3:
(You left this blank)
Biological Species Concept
Ecological Species Concept
Phylogenetic Species Concept
Genotypic Cluster Species Concept
Taxonomic Species

Answer 4:
(You left this blank)
Biological Species Concept
Ecological Species Concept
Phylogenetic Species Concept
Genotypic Cluster Species Concept
Taxonomic Species

Answer 5:
(You left this blank)
Biological Species Concept
Ecological Species Concept
Phylogenetic Species Concept
Genotypic Cluster Species Concept
Taxonomic Species

Answer 6:
(You left this blank)
Biological Species Concept
Ecological Species Concept

Phylogenetic Species Concept

Genotypic Cluster Species Concept

Taxonomic Species

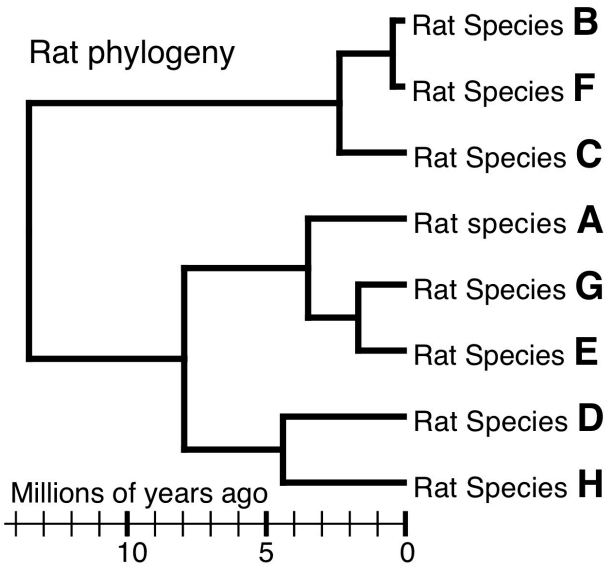
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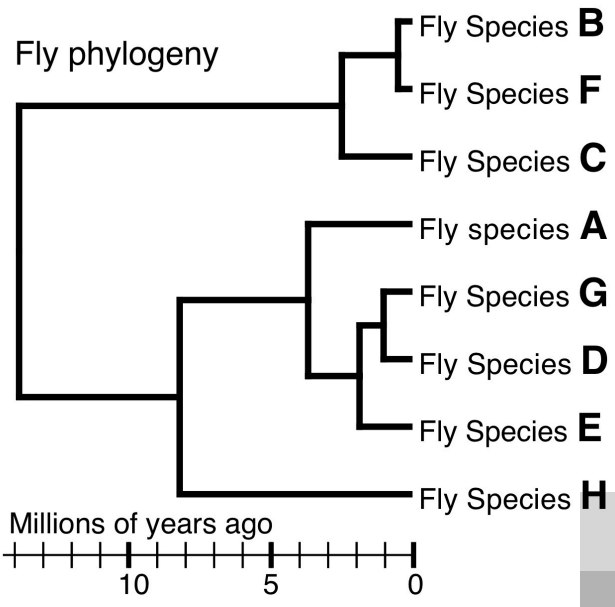
Question 19

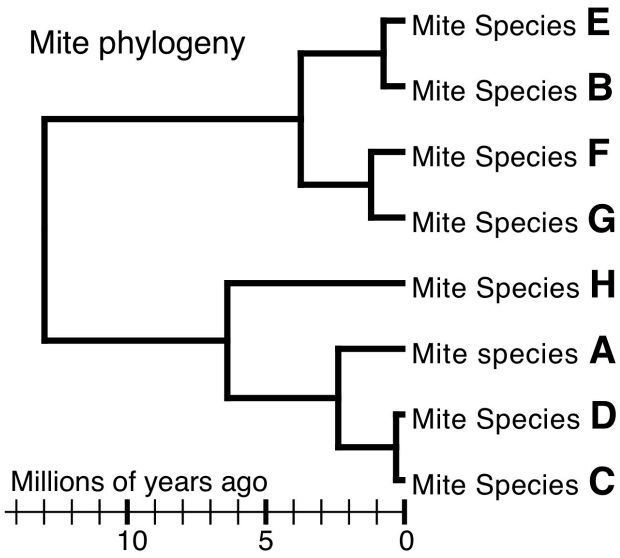
0 / 3 pts

(This question does not apply this term)

A group of 8 species of rats named A, B, C, D, E, F, G, and H are discovered to have parasites. Some are flightless flies, and they form a group of 8 species. Each species of fly is found on one species of rats (fly species A is found on rat species A, fly B on rat B, etc.). Other parasites are mites, and they form a group of 8 species. Each species of mite is found on one species of rats (mite species A is found on rat species A, mite B on rat B, etc.). Here are the phylogenies of the rats, flies, and mites.







According to the principles discussed in lecture, which group of parasites is most likely to be vertically inherited? [Select]

Which parasites are more likely to lead to dangerous health complications for the rats? [Select]

In one of these two groups there is evidence for one shift of a parasite lineage from one lineage of rats to another lineage of rats. Approximately how long ago did it occur? [Select]

Answer 1:
(You left this blank)
fly
mite
Answer 2:
(You left this blank)
fly
mite
Answer 3:
(You left this blank)
0 to 3 million years ago
3 to 6 million years ago
6 to 9 million years ago
more than 9 million years ago
Unanswered

Question 20
0 / 3 pts

[ADDED FOR THIS TERM INSTEAD OF PREVIOUS QUESTION]

In lab studies, symbionts that are forced to transmit [Select] often evolve to become more beneficial. This occurs because this type of transmission leads to interactions that are more like the iterated prisoner's dilemma where partners [Select] , as they remain within the same host. In addition, symbionts living within a host are more [Select] .

Answer 1:
(You left this blank)
vertically
horizontally
obliquely
Answer 2:
(You left this blank)
are mixed at every interaction
remain paired across multiple interactions
Answer 3:
(You left this blank)
related (increasing r in rB>C of Hamilton's rule)
important (increasing B in rB>C of Hamilton's rule)
virulent (increasing C in rB>C of Hamilton's rule)